

Unit I- Introduction to Database Management System

Syllabus: Introduction to Database Management Systems, Purpose of Database Systems, Database-System Applications, View of Data, Database Languages, Database System Structure, Enterprise Constraints Data Models, Database Design and ER Model: Entity, Attributes, Relationships, Constraints, Keys, Design Process, Entity Relationship Model, ER Diagram, Design Issues, Extended E-R Features, Converting E-R & EER diagram into tables

1. Introduction to Database Management Systems / Purpose of Database Systems

Topics include: Purpose of Database Systems, traditional file system differences, and limitations.

Q1) List the main characteristics of the database approach and explain how it differs from the traditional file system.

Main Characteristics of Database Approach

1. Self-Describing Nature

A DBMS stores:

- Data
- Structure of data (Metadata)

Example: Table names, column names, data types.

2. Program and Data Independence

Programs are independent of data structure.

Changes in database structure do not require major changes in application programs.

3. Data Abstraction

Users can access only required data without knowing storage details.

Levels:

- Physical Level
- Logical Level
- View Level

4. Multiple User Support

Many users can access the database at the same time.

Example:

Bank customers using ATM simultaneously.

5. Data Sharing

Data can be shared among different users and applications.

6. Reduced Data Redundancy

Duplicate data is minimized.

This saves storage space and improves consistency.

7. Data Integrity

DBMS maintains accuracy and consistency of data using constraints.

Example:

Primary key cannot be NULL.

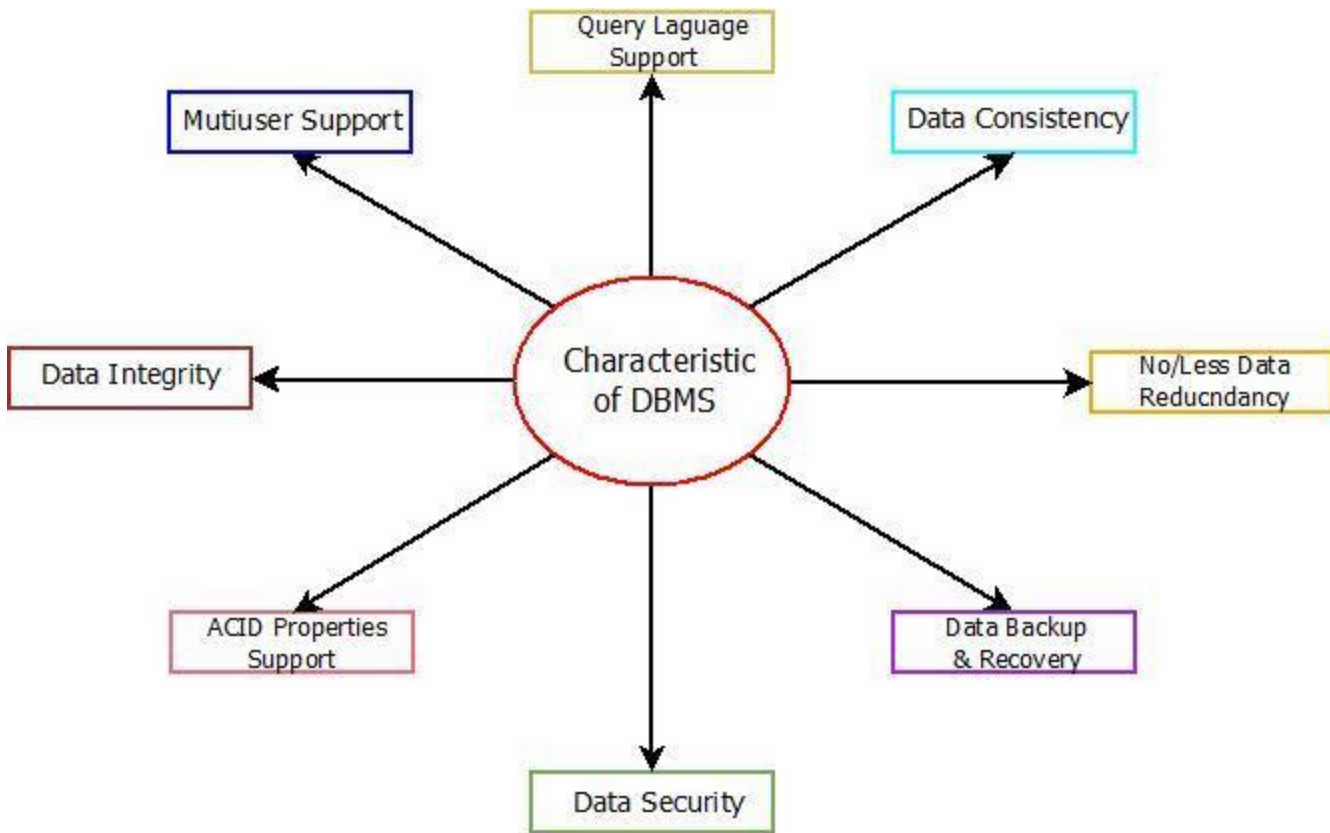
8. Security

DBMS provides authorization and access control.

Only authorized users can access data.

9. Backup and Recovery

DBMS provides backup and recovery facilities in case of system failure.



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Difference Between Database Approach and Traditional File System

Database Approach (DBMS)	Traditional File System
Data redundancy is reduced	High data redundancy
Better data consistency	Data inconsistency possible
High security	Low security
Supports multiple users	Limited multi-user support
Data sharing is easy	Difficult data sharing
Backup and recovery available	Poor backup facility
Centralized control	Separate files managed independently
Data independence present	No data independence
Easy querying using SQL	Difficult data retrieval

Conclusion

The database approach is more efficient, secure, and reliable than the traditional file system because it reduces redundancy, improves consistency, supports multiple users, and provides better security and recovery mechanisms.

Q2) Compare DBMS and File Processing System

Point	DBMS	File Processing System
Data Isolation	Data is stored together and easily accessible.	Data is stored in separate files, difficult to access together.
Data Redundancy	Duplicate data is reduced.	Same data is repeated in many files.
Data Inconsistency	Data remains accurate and consistent.	Different files may contain different values of same data.
Data Integrity	Rules and constraints maintain correct data.	No proper control on data accuracy.
Security	Provides better security and access control.	Security is less.
Backup and Recovery	Backup and recovery facilities are available.	Backup and recovery are difficult.

Conclusion

DBMS is better than File Processing System because it reduces redundancy, improves consistency, provides security, and maintains data integrity.

2. View of Data

Topics include: Levels of data abstraction and data independence.

View of Data

View of Data means how data is seen by users in a database system.
DBMS hides unnecessary details from users using different levels of abstraction.

Levels of Data Abstraction

There are mainly 3 levels of data abstraction:

1. Physical Level

- Lowest level of abstraction.
- Describes how data is stored in memory.
- Includes storage details like files and indexes.

Example:

Data stored in hard disk blocks.

2. Logical Level

- Describes what data is stored in database.
- Defines relationships among data.

Example:

Student table contains:

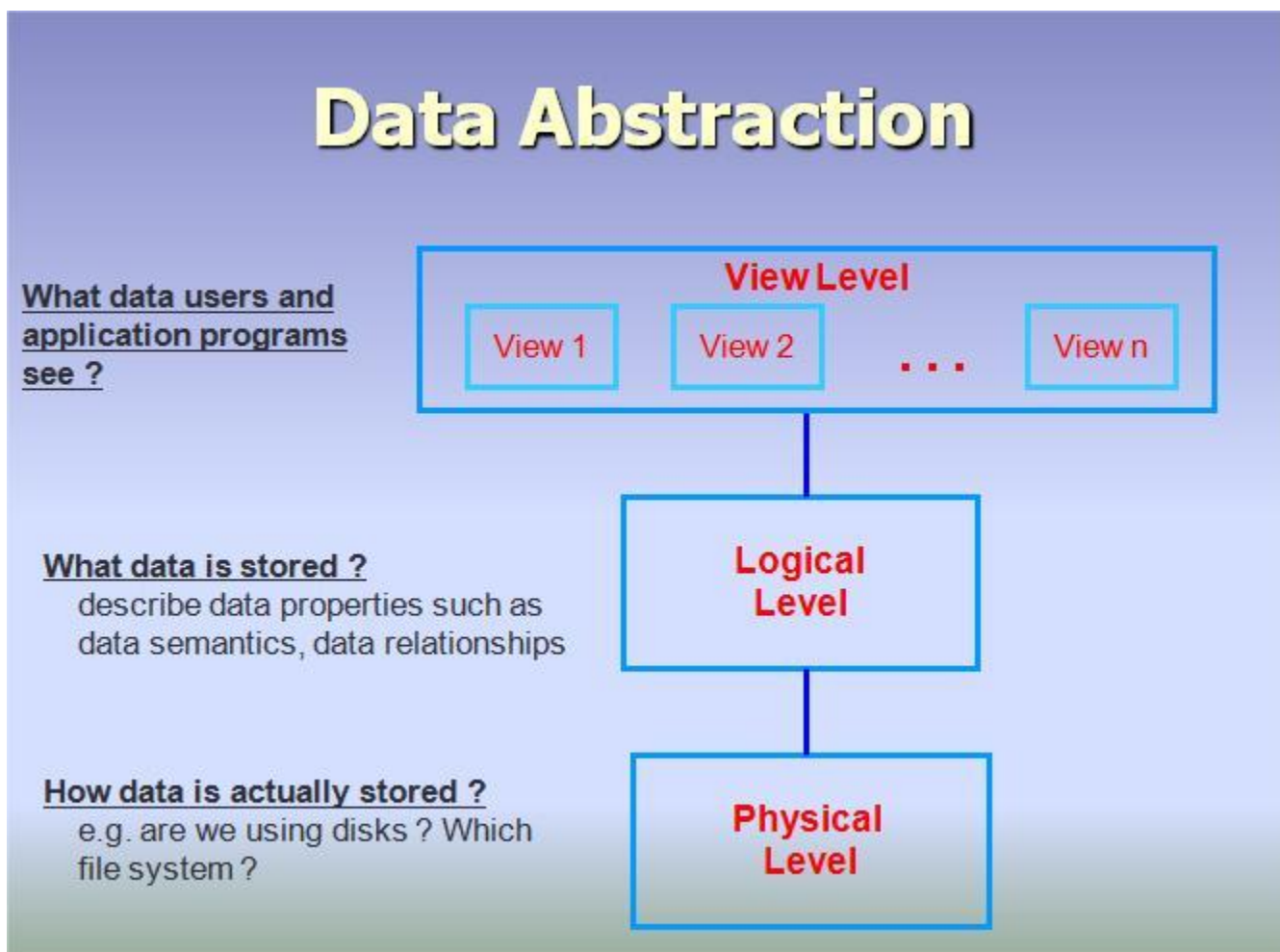
- Roll No
- Name
- Marks

3. View Level

- Highest level of abstraction.
- Users can see only required data.
- Provides security by hiding unnecessary information.

Example:

Student can view only his marks, not all students' records.



Data Independence

Data Independence means changes in database structure do not affect application programs.

Types of Data Independence

1. Physical Data Independence

- Changes in physical storage do not affect logical structure.

Example:

Changing storage device from HDD to SSD.

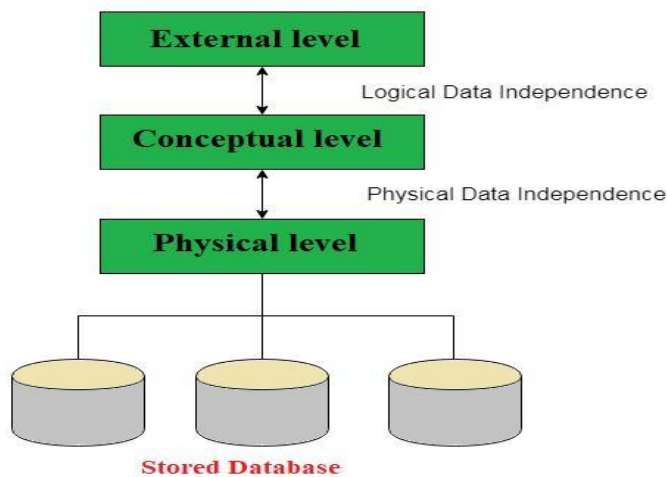
2. Logical Data Independence

- Changes in logical structure do not affect user views.

Example:

Adding a new column in table without changing application.

Data Independence in DBMS



Advantages of Data Abstraction and Independence

1. Improves security
2. Easy database maintenance
3. Reduces complexity
4. Easier application development
5. Better data management

Short Note

- **Physical Level** → How data is stored
- **Logical Level** → What data is stored
- **View Level** → What user sees
- **Physical Data Independence** → Storage changes do not affect programs
- **Logical Data Independence** → Logical changes do not affect users

Q3. For the database system to be usable, it must retrieve data efficiently. The need of efficiency has led designers to use complex data structures to represent data in the database. Developers hide this complexity from the database system users through several levels of abstraction. Explain those levels of abstraction in detail with example. [5]

In a database system, data is stored using complex structures.

To make the system easy for users, DBMS hides these details using different levels of abstraction.

Levels of abstraction help:

- Reduce complexity
 - Improve security
 - Make database easy to use
-

Levels of Abstraction in DBMS

There are three levels of abstraction:

1. Physical Level (Lowest Level)

- Describes how data is physically stored in memory.
- Deals with storage devices, files, indexes, etc.
- Managed by database administrators.

Example:

Student records stored in hard disk blocks.

2. Logical Level (Middle Level)

- Describes what data is stored in database.
- Defines tables, attributes, and relationships.

Example:

Student Table:

RollNo	Name	Marks
--------	------	-------

101	Amit	85
-----	------	----

Here, user knows structure of table but not physical storage details.

3. View Level (Highest Level)

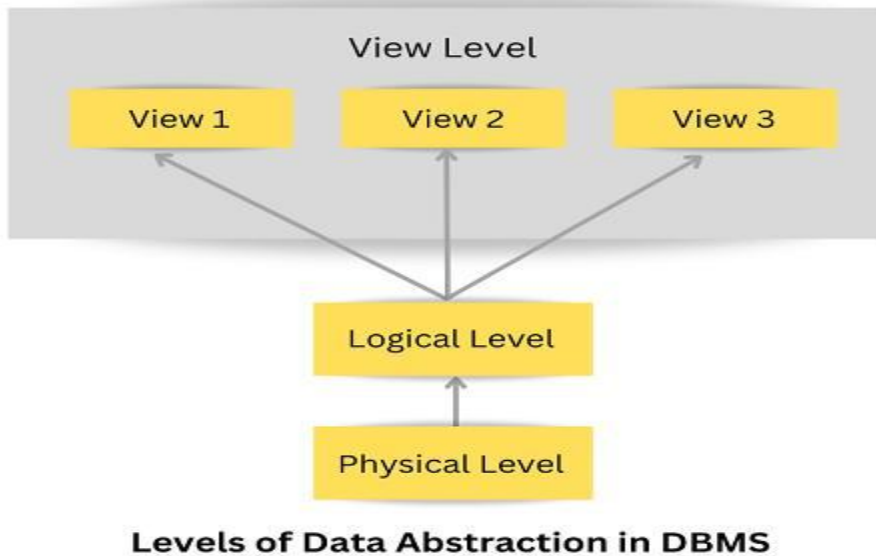
- Describes only part of database required by users.
- Different users can have different views.
- Provides data security.

Example:

A student can view only:

- Name
- Marks

But cannot see salary details of staff.



Example of Levels of Abstraction

Suppose a college database contains:

- Student details
- Teacher details
- Fees information

Physical Level

Data stored in hard disk sectors.

Logical Level

Tables like:

- Student(RollNo, Name, Marks)
- Teacher(ID, Name, Subject)

View Level

Student sees only his result information.

Advantages of Data Abstraction

1. Reduces complexity
2. Improves security
3. Easy database maintenance
4. Easy data access
5. Hides internal details

Levels of abstraction help users interact with database systems easily by hiding complex storage details and showing only required information.

Q4) Explain with example what is physical data independence. Also explain its importance. [4]

Physical data independence means changes made in the physical storage of the database do not affect the application programs or users.

It allows modification in storage methods without changing the logical structure of the database.

Physical data independence helps:

- Improve database performance
- Reduce maintenance work
- Provide flexibility in storage
- Make system easier to manage

Physical Data Independence

Physical data independence is the ability to change physical level without affecting logical level.

Changes can include:

- Changing storage devices
- Changing file organization
- Creating or deleting indexes
- Changing data storage location

These changes do not affect:

- Tables
- Application programs
- User operations

Example of Physical Data Independence

Suppose a college database stores student records on a hard disk.

Initially:

- Data is stored using normal file storage.

Later:

- Database administrator changes storage to SSD for faster access.
- Indexes are created to improve performance.

Even after these changes:

- Student table structure remains same.
- Application programs continue working normally.
- Users do not notice any change.

Example Table:

RollNo	Name	Marks
101	Amit	85
102	Neha	90

Only physical storage changes, not table structure.

Importance of Physical Data Independence

1. Improves system performance
2. Allows changes in storage without affecting users
3. Reduces application modification
4. Makes database maintenance easier
5. Increases flexibility of database system

Physical data independence helps DBMS improve storage methods and performance without changing application programs or affecting users.

Database Languages

Database languages are used to create, manage, manipulate, and control databases in DBMS.

Types of Database Languages

1. DDL (Data Definition Language)

DDL is used to define database structure.

Commands:

- CREATE
- ALTER
- DROP
- TRUNCATE

Example:

```
CREATE TABLE Student (  
    RollNo INT,
```

```
    Name VARCHAR(20)
);
```

2. DML (Data Manipulation Language)

DML is used to insert, update, and delete data from database.

Commands:

- INSERT
- UPDATE
- DELETE

Example:

```
INSERT INTO Student
VALUES (101, 'Amit');
```

3. DQL (Data Query Language)

DQL is used to retrieve data from database.

Command:

- SELECT

Example:

```
SELECT * FROM Student;
```

4. DCL (Data Control Language)

DCL is used to control user permissions and security.

Commands:

- GRANT
- REVOKE

Example:

```
GRANT SELECT
ON Student TO User1;
```

5. TCL (Transaction Control Language)

TCL is used to manage transactions in database.

Commands:

- COMMIT
- ROLLBACK

- SAVEPOINT

Example:

```
COMMIT;
```

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Summary Table

Language	Purpose	Examples
DDL	Defines database structure	CREATE, ALTER
DML	Manipulates data	INSERT, UPDATE
DQL	Retrieves data	SELECT
DCL	Controls access	GRANT, REVOKE
TCL	Manages transactions	COMMIT, ROLLBACK

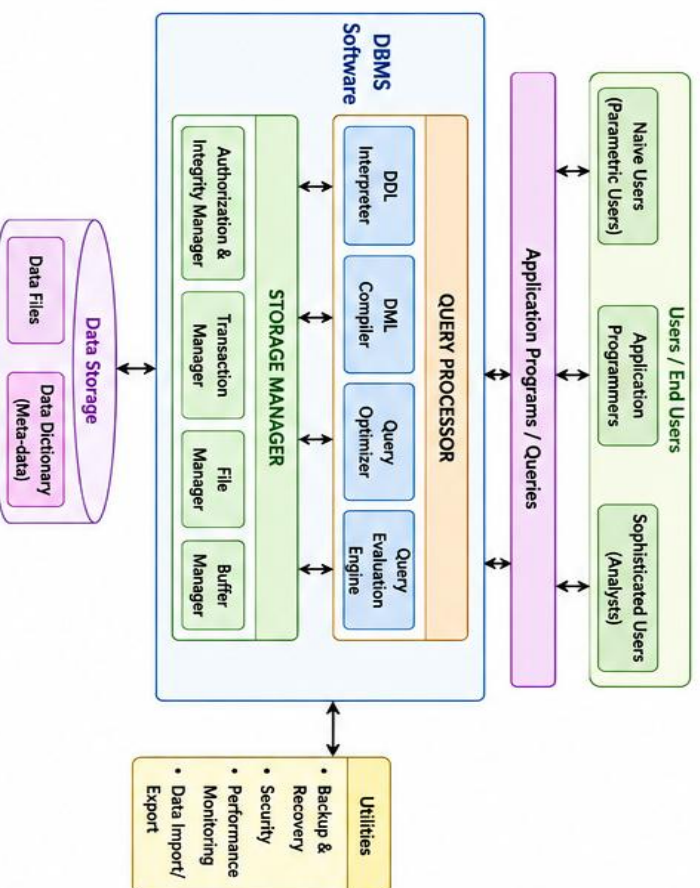
Conclusion

Database languages help users create, manage, retrieve, and secure data efficiently in a database system.

Database System Structure:

Q4) Draw architecture of DBMS system and explain function of following components: [9]

DBMS Architecture



i) STORAGE MANAGER

The storage manager is responsible for the interaction between the low level stored data and the application programs and queries submitted to the system.

Functions:

1. **Authorization and Integrity Manager** – Checks user authorization for accessing data and ensures integrity constraints.
2. **Transaction Manager** – Manages concurrent transactions, ensures ACID properties (Atomicity, Consistency, Isolation, Durability).
3. **File Manager** – Manages file organization, space allocation on disk and access paths.
4. **Buffer Manager** – Handles data transfer between main memory and disk to improve performance.

ii) QUERY PROCESSOR

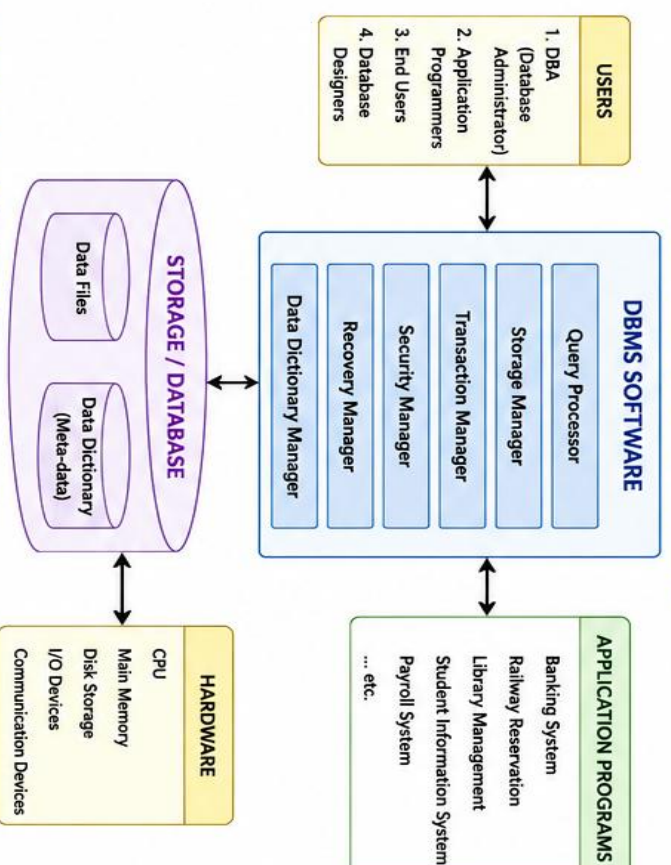
The query processor interprets the queries and converts them into efficient instructions that the storage manager can execute.

Functions:

1. **DDL Interpreter** – Interprets DDL statements (CREATE, ALTER, DROP etc.) and records the definitions in the data dictionary.
2. **DML Compiler** – Converts DML queries into low level code (execution plan).
3. **Query Optimizer** – Chooses the most efficient execution plan for a query.
4. **Query Evaluation Engine** – Executes the optimized plan and returns the result to users.

Q8) Draw the neat diagram of Database System Structure and explain its components in detail. [8].

Database System Structure



Components and Explanation

1. **Users:** People who interact with the database system.
 - **DBA** – Overall control of database, defines schema, authorizes users, handles backup & recovery.
 - **Application Programmers** – Write programs to access the database.
 - **End Users** – Access data through application programs (naive, casual, sophisticated users).
 - **Database Designers** – Design the database schema and structure.
2. **DBMS Software:** It is the software that manages the entire database system.
 - **Query Processor** – Interprets and processes user queries.
 - **Storage Manager** – Manages storage of data in memory and on disk.
 - **Transaction Manager** – Controls concurrent transactions ensuring ACID properties.
 - **Security Manager** – Checks authorization and enforces security.
 - **Recovery Manager** – Ensures recovery in case of failures.
 - **Data Dictionary Manager** – Maintains metadata (data about data).
3. **Storage / Database:** Contains the actual data of the organization.
 - **Data Files** – Store the actual user data.
 - **Data Dictionary (Meta-data)** – Stores structure of data, constraints, schema, etc.
4. **Application Programs:** Programs written to perform a specific task by accessing database. Examples: Banking system, Library system, Payroll system, etc.
5. **Hardware:** Provides the physical resources required to store and process data. Examples: CPU, Main Memory, Disk, I/O Devices, Communication Devices.

Enterprise Constraints

Enterprise constraints are rules applied on a database to maintain accuracy, consistency, and validity of data in an organization.

These constraints ensure that wrong or invalid data cannot be entered into the database.

Types of Enterprise Constraints

1. Domain Constraint

- Defines valid values for an attribute.
- Data must belong to a specific domain.

Example:

Age must be between 18 and 60.

2. Key Constraint

- Ensures uniqueness of data.

Example:

Roll Number of students must be unique.

3. Entity Integrity Constraint

- Primary key cannot be NULL.

Example:

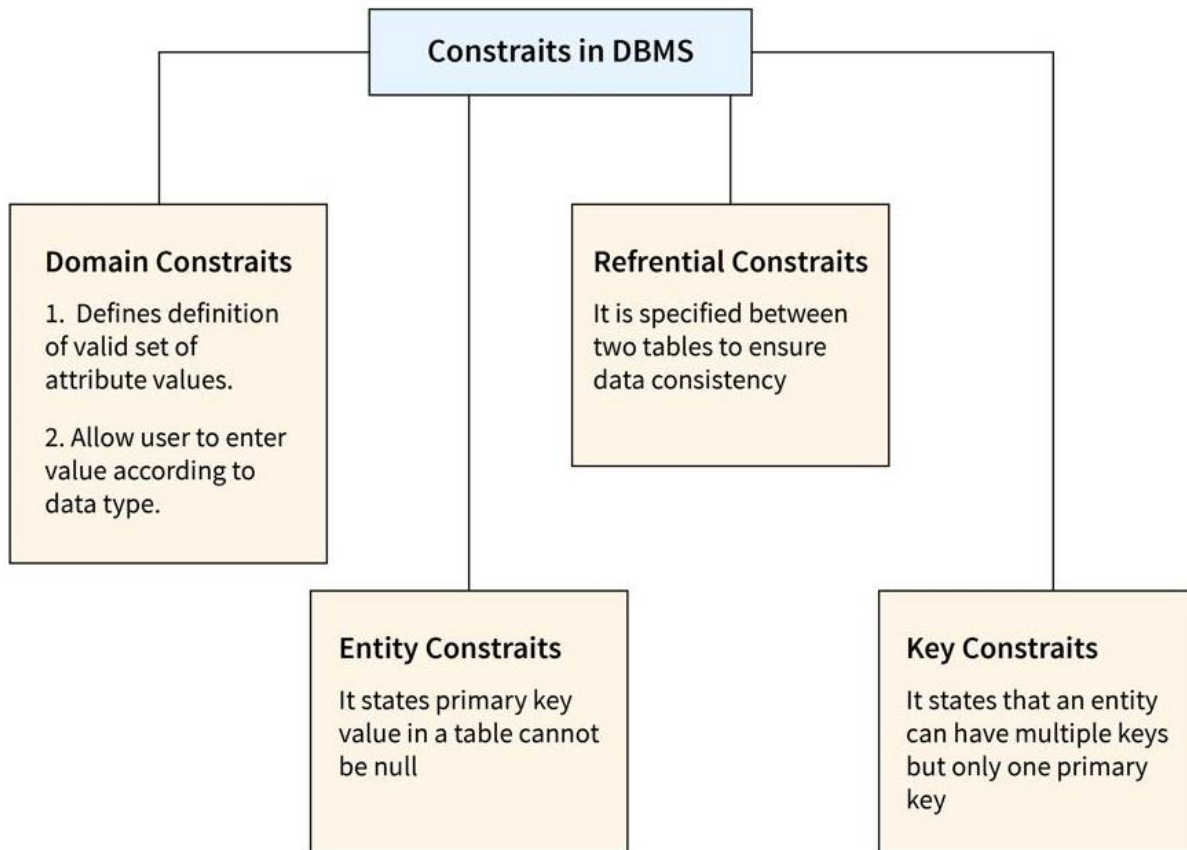
StudentID cannot be empty.

4. Referential Integrity Constraint

- Foreign key value must match primary key value of another table.

Example:

CourseID in Student table must exist in Course table.



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Advantages of Enterprise Constraints

1. Maintains data accuracy
2. Prevents invalid data entry
3. Improves data consistency
4. Maintains relationships between tables
5. Enhances database reliability

Data Models

A Data Model is a collection of concepts used to describe the structure of a database.

It defines:

- How data is stored
 - Relationship among data
 - Constraints on data
-

Types of Data Models

1. Hierarchical Data Model

- Data organized in tree structure.
- One parent can have many children.

Example:

Company → Department → Employee

2. Network Data Model

- Data organized using links.
- One record can have multiple parent records.

Example:

Student connected to multiple courses.

3. Relational Data Model

- Data stored in tables (rows and columns).
- Most widely used model.

Example:

Student Table

RollNo	Name	Marks
--------	------	-------

4. Object-Oriented Data Model

- Data represented as objects.
- Supports classes and inheritance.

Example:

Student object containing:

- Name
- RollNo
- Methods

Advantages of Data Models

1. Easy database design
 2. Better data organization
 3. Improves data consistency
 4. Easy relationship management
 5. Simplifies database development
-

Conclusion

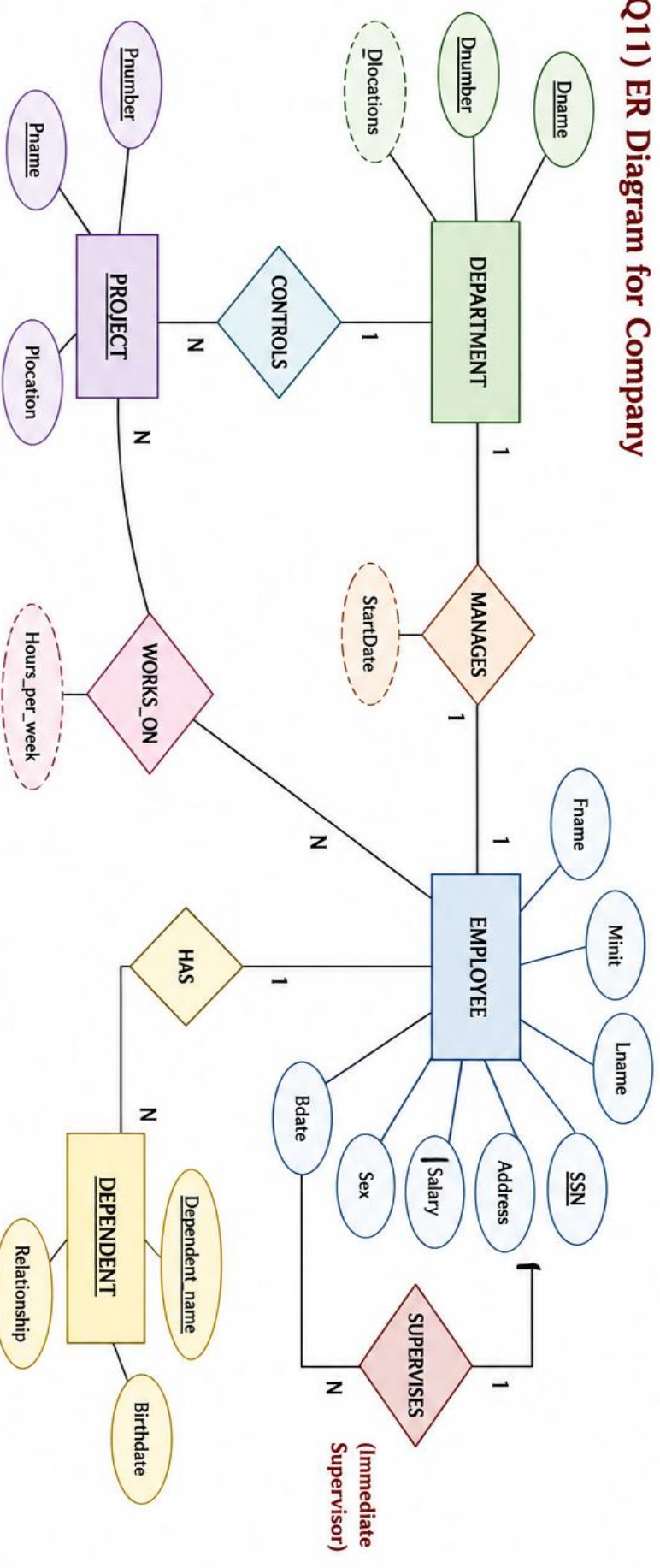
Enterprise constraints maintain correctness of data, while data models define how data is organized and stored in a database system.

Database Design and ER Model: Entity, Attributes, Relationships, Constraints

Q7) Construct an ER Diagram for Company having following details : [7]

- Company organized into DEPARTMENT. Each department has unique name and a particular employee who manages the department. Start date for the manager is recorded. Department may have several locations.
- A department controls a number of PROJECT. Projects have a unique name, number and a single location.
- Company's EMPLOYEE name, ssno, address, salary, sex and birth date are recorded. An employee is assigned to one department, but may work for several projects (not necessarily controlled by her dept). Number of hours/week an employee works on each project is recorded; The immediate supervisor for the employee.
- Employee's DEPENDENT are tracked for health insurance purposes (dependent name, birthdate, relationship to employee).
- Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram.

Q11) ER Diagram for Company



ENTITIES AND KEYS

DEPARTMENT (Dnumber, Dname, Dlocations)
 Key : Dnumber

EMPLOYEE (SSN, Fname, Minit, Lname, Address, Salary, Sex, Bdate)
 Key : SSN

PROJECT (Pnumber, Pname, Plocation)
 Key : Pnumber

DEPENDENT (Dependent_name, Birthdate, Relationship, SSN*)
 Key : (Dependent_name, SSN)

SSN* : Foreign Key referencing EMPLOYEE(SSN)

RELATIONSHIPS, MAPPING CARDINALITIES & ATTRIBUTES

- MANAGES** : EMPLOYEE manages DEPARTMENT
 Mapping : 1 : 1
 Attribute : StartDate (start date of the manager)
- CONTROLS** : DEPARTMENT controls PROJECT
 Mapping : 1 : N (One department controls many projects)
- WORKS_ON** : EMPLOYEE works on PROJECT
 Mapping : M : N (An employee may work on several projects and a project can have many employees)
 Attribute : Hours_per_week (number of hours/week)
- SUPERVISES** : EMPLOYEE supervises EMPLOYEE
 Mapping : 1 : N (One supervisor can supervise many employees) (Immediate Supervisor)
- HAS** : EMPLOYEE has DEPENDENT
 Mapping : 1 : N (An employee can have many dependents)

CONSTRAINTS / BUSINESS RULES

- Each department has unique name.
- Each department is managed by one employee.
- Each department may have several locations.
- Each project has unique name, number and a single location.
- Each employee is assigned to one department.
- An employee may work for several projects (not necessarily controlled by her department).
- Number of hours per week is recorded for each employee-project pair.
- Each employee has an immediate supervisor (another employee).
- Dependents are tracked for each employee for insurance purposes.

Strong Entity

Weak Entity

Relationship

Attribute

Multivalued Attribute

Underlined Attribute : Key Attribute

Keys

8) Explain the concept of candidate key and primary key, foreign key. Identify above listed key for the following schema: [6]

Person (driver_id, name, address, contactno) Car(licence, model, year) Owns (driver_id, licence)

a) Candidate Key, Primary Key and Foreign Key

1. Candidate Key

A candidate key is an attribute or set of attributes that can uniquely identify a record in a table.

A table can have multiple candidate keys.

Example:

In STUDENT(RollNo, Email, Name)

- RollNo
- Email

Both can uniquely identify student records, so both are candidate keys.

2. Primary Key

A primary key is the candidate key selected to uniquely identify records in a table.

Properties:

- Unique
- Cannot be NULL

Example:

RollNo chosen as primary key.

3. Foreign Key

A foreign key is an attribute that creates relationship between two tables.

It refers to the primary key of another table.

Given Schema

Person (driver_id, name, address, contactno)

Car (licence, model, year)

Owns (driver_id, licence)

Identification of Keys

Table	Candidate Key	Primary Key	Foreign Key
Person	driver_id	driver_id	—
Car	licence	licence	—
Owens	(driver_id, licence)	(driver_id, licence)	driver_id references Person(driver_id), licence references Car(licence)

Explanation

Person Table

- driver_id uniquely identifies each person.
 - So it is both candidate key and primary key.
-

Car Table

- licence uniquely identifies each car.
 - So it is both candidate key and primary key.
-

Owens Table

- One person may own many cars.
- One car belongs to one owner.

Combination of:

- driver_id
- licence

uniquely identifies records in Owens table.

Hence:

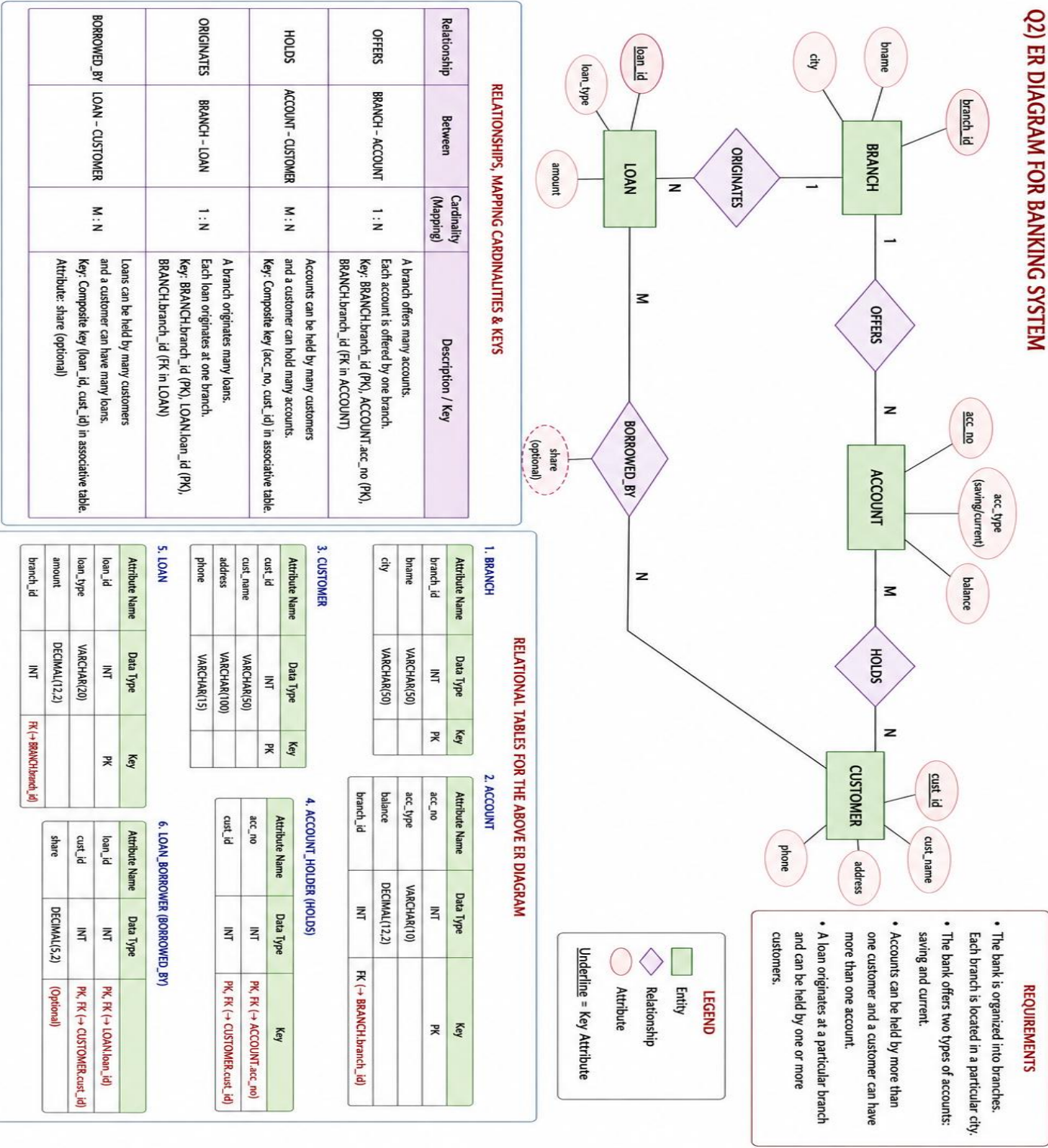
- Composite Primary Key = (driver_id, licence)
- Both are also foreign keys.

Design Process :

Entity Relationship Model / ER Diagram

Q2) Draw an ER diagram for the banking system. Assume the banking requirements are as given below. [10]

- The bank is organized into branches. Each branch is located in a particular city.
- The bank offers two types of accounts: saving and current. Accounts can be held by more than one customer and a customer can have more than one account.
- A loan originates at a particular branch and can be held by one or more customers
- Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram. Construct appropriate tables for E-R diagram designed with above requirements.

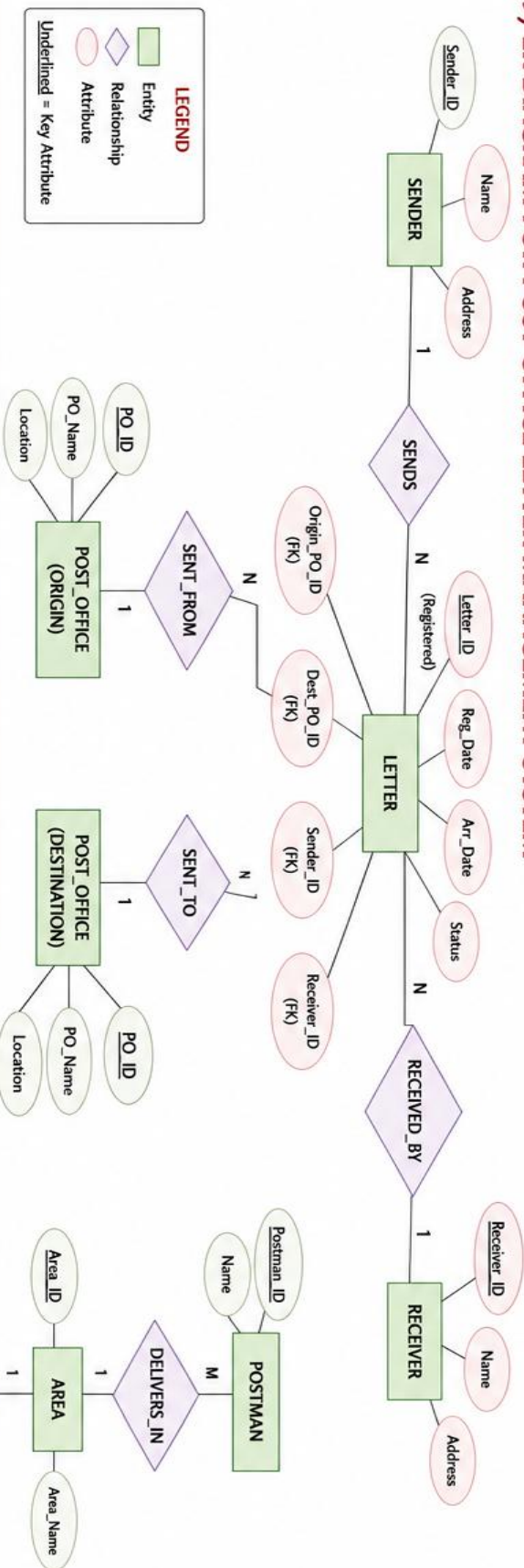


Q9) A post office has few postmen who go every day to distribute letter. Every morning post office receives a large number of registered letters. the post office intends to create a database to keep track of these letters. [7]

- i) Every letter has a sender, an origin post office from where it was sent, a destination post office to which it is to be sent, a date of registration, date of arrival at destination post office, receiver and a status.
- ii) Every sender has a name, an address.
- iii) Every receiver has a name, an address
- iv) Every postman has a designated area where he delivers letters.
- v) The area consists of a set of streets under the jurisdiction of the post office.
- vi) Every street consists of a set of buildings.
- vii) Every building has number and may be name. It may be housing more than one family.
- viii) The status of the letter can be not yet taken for delivery, delivered, address not available, address not known, addressee did not accept the letter, redirected to the address of address and sent to the sender.
- Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram. Construct appropriate tables for E-R diagram designed with above requirements

Q15) A reputed general hospital has decided to computerize their operations. In the hospital many doctors are working. Personal information of doctors is maintained to get them fixed salary per month. The patients are admitted to the hospital into the room. They are treated by various doctors. Sometimes patients perform certain pathological tests which carried out into the labs. Identify the relationship among the entities along with the mapping cardinalities, keys in the E.R. diagram. Construct appropriate tables for E-R diagram designed with above requirements. [7].

Q9) ER DIAGRAM FOR POST OFFICE LETTER MANAGEMENT SYSTEM



RELATIONAL TABLES FOR THE ABOVE ER DIAGRAM

1. SENDER

Field Name	Data Type	Key
Sender_ID	INT	PK
Name	VARCHAR(100)	
Address	VARCHAR(200)	

2. RECEIVER

Field Name	Data Type	Key
Receiver_ID	INT	PK
Name	VARCHAR(100)	
Address	VARCHAR(200)	

3. POST_OFFICE

Field Name	Data Type	Key
PO_ID	INT	PK
PO_Name	VARCHAR(100)	
Location	VARCHAR(100)	

4. LETTER

Field Name	Data Type	Key
Letter_ID	INT	PK
Sender_ID	INT	FK ↔ SENDER(Sender_ID)
Receiver_ID	INT	FK ↔ RECEIVER(Receiver_ID)
Origin_PO_ID	INT	FK ↔ POST_OFFICE(PO_ID)
Dest_PO_ID	INT	FK ↔ POST_OFFICE(PO_ID)
Reg_Date	DATE	
Arr_Date	DATE	
Status	VARCHAR(50)	

5. POSTMAN

Field Name	Data Type	Key
Postman_ID	INT	PK
Name	VARCHAR(100)	
Area_ID	INT	FK ↔ AREA(Area_ID)

6. AREA

Field Name	Data Type	Key
Area_ID	INT	PK
Area_Name	VARCHAR(100)	

7. STREET

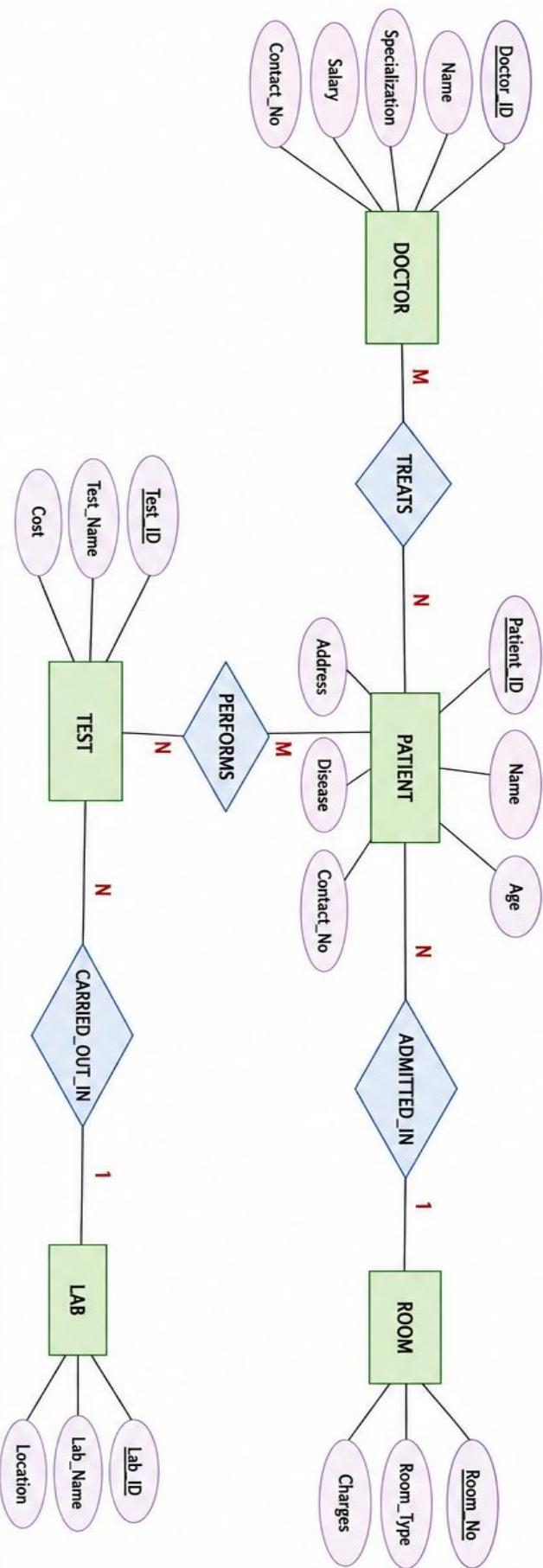
Field Name	Data Type	Key
Street_ID	INT	PK
Street_Name	VARCHAR(100)	
Area_ID	INT	FK ↔ AREA(Area_ID)

8. BUILDING

Field Name	Data Type	Key
Building_ID	INT	PK
Building_No	VARCHAR(20)	
Building_Name	VARCHAR(100)	
Street_ID	INT	FK ↔ STREET(Street_ID)

Q15) ER DIAGRAM FOR HOSPITAL MANAGEMENT SYSTEM

A reputed general hospital has decided to computerize their operations. In the hospital many doctors are working. Personal information of doctors is maintained to get them fixed salary per month. The patients are admitted to the hospital in the room. They are treated by various doctors. Sometimes patients perform certain pathological tests which carried out in the labs.



RELATIONSHIPS, MAPPING CARDINALITIES & KEYS

Relationship	Between	Cardinality (Mapping)	Description / Key
TREATS	DOCTOR – PATIENT	M : N	One doctor can treat many patients and one patient can be treated by many doctors. Key: Doctor_ID (FK), Patient_ID (FK)
ADMITTED_IN	PATIENT – ROOM	N : 1	Many patients can be admitted in one room but one patient in one room at a time. Key: Room_No (FK) in PATIENT
PERFORMS	PATIENT – TEST	M : N	One patient may perform many tests and one test may be performed for many patients. Key: Patient_ID (FK), Test_ID (FK)
CARRIED_OUT_IN	TEST – LAB	N : 1	Many tests are carried out in one lab, but one test belongs to one lab. Key: Lab_ID (FK) in TEST

RELATIONAL TABLES FOR THE ABOVE ER DIAGRAM

1. DOCTOR

Attribute Name	Data Type	Key
Doctor_ID	INT	PK
Name	VARCHAR(100)	
Specialization	VARCHAR(100)	
Salary	DECIMAL(10,2)	
Contact_No	VARCHAR(15)	

2. PATIENT

Attribute Name	Data Type	Key
Patient_ID	INT	PK
Name	VARCHAR(100)	
Age	INT	
Address	VARCHAR(200)	
Disease	VARCHAR(100)	
Contact_No	VARCHAR(15)	
Room_No	INT	FK (→ ROOM.Room_No)

3. ROOM

Attribute Name	Data Type	Key
Room_No	INT	PK
Room_Type	VARCHAR(50)	
Charges	DECIMAL(10,2)	

4. LAB

Attribute Name	Data Type	Key
Lab_ID	INT	PK
Lab_Name	VARCHAR(100)	
Location	VARCHAR(100)	

5. TEST

Attribute Name	Data Type	Key
Test_ID	INT	PK
Test_Name	VARCHAR(100)	
Cost	DECIMAL(8,2)	
Lab_ID	INT	FK (→ Lab_ID)

6. TREATS (Associative Table)

Attribute Name	Data Type	Key
Doctor_ID	INT	PK, FK (→ DOCTOR.Doctor_ID)
Patient_ID	INT	PK, FK (→ PATIENT.Patient_ID)

7. PERFORMS (Associative Table)

Attribute Name	Data Type	Key
Patient_ID	INT	PK, FK (→ PATIENT.Patient_ID)
Test_ID	INT	PK, FK (→ TEST.Test_ID)

PK – Primary Key FK – Foreign Key

Design Issues

- **Q16)** A weak entity set can always be made into a strong entity set ; by adding to its attributes the primary key attributes of its identifying entity set. Outline what sort of redundancy will result if we do so while converting to tables. **[4]**.

Extended E-R Features

Specialization and Generalization in Extended E-R Diagram

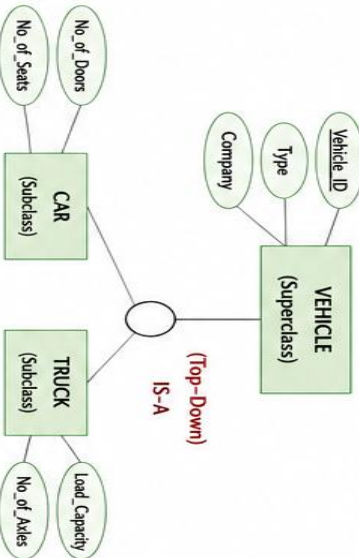
Specialization and generalization are used to model inheritance (IS-A relationship) between entity sets in Extended E-R model.

3. DIFFERENCE BETWEEN SPECIALIZATION AND GENERALIZATION

Basis	Specialization	Generalization
1. Direction	Top-Down (Superclass → Subclass)	Bottom-Up (Subclass → Superclass)
2. Starting Point	Starts with a higher level entity (superclass)	Starts with lower level entities (subclasses)
3. Result	Creates multiple subclasses	Creates a single superclass
4. Example	VEHICLE → CAR, TRUCK	CAR, TRUCK → VEHICLE
5. Purpose	To divide a general entity into specific entities	To combine similar entities into a general entity

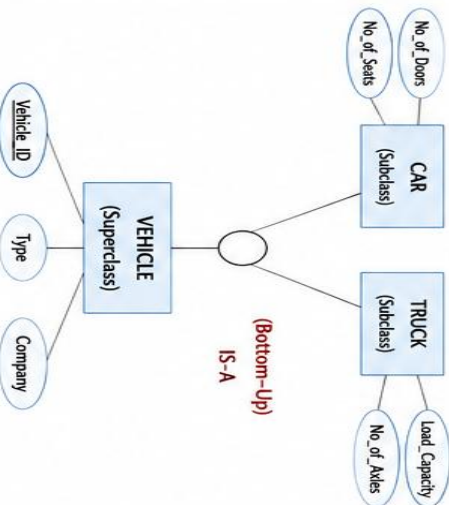
1. SPECIALIZATION (Top-Down Approach)

- Specialization is the process of deriving lower level (subclass) entity sets from a higher level (superclass) entity set.
- The superclass contains common attributes which are inherited by its subclasses.
- It represents a top-down approach.



2. GENERALIZATION (Bottom-Up Approach)

- Generalization is the process of deriving a higher level (superclass) entity set from lower level (subclass) entity sets.
- It represents a bottom-up approach.
- Common attributes of subclasses are placed in the superclass.



4. WHY DO WE NOT DISPLAY THE DIFFERENCE IN SCHEMA DIAGRAM?

- In the relational schema, specialization and generalization are implemented in the same way.
- The same set of tables (relations) are created whether we use specialization or generalization.
- Only the direction of thinking is different, but the resulting relational tables remain same.
- Therefore, schema diagram does not show any difference between them.

Example Schema (Same for Specialization and Generalization)

VEHICLE	CAR	TRUCK
Vehicle_ID (PK)	Vehicle_ID (PK, FK)	Vehicle_ID (PK, FK)
Type	No. of Doors	Load Capacity
Company	No. of Seats	No. of Axles

CAR and TRUCK inherit Vehicle_ID from VEHICLE. Vehicle_ID in CAR and TRUCK is both Primary Key and Foreign Key referencing VEHICLE.

5. SUMMARY

- Specialization: Superclass to Subclass (Top-Down)
- Generalization: Subclass to Superclass (Bottom-Up)
- Both represent IS-A relationship.
- Implementation in relational schema is same.
- Hence, schema diagram does not display any difference.

LEGEND

Entity Set	Attribute	Specialization / Generalization (ISA)	Connection

Extended E-R Features

Q6) What is Specialization and generalization in Extended E-R diagram? What is the difference between them? Why do we not display the difference in schema diagram? [6].

Converting E-R & EER diagram into tables